

CHAPTER 5 GAS VAPOR MIXTURE AND AIR CONDITIONING

- We looked at mixture of gases (mass fractions, mole fractions, P-v-T , properties) where all components in the mixture remained in the gas phase.
- We now consider the case where one of the components can go through a phase change.

DRY AND ATMOSPHERIC AIR

Atmospheric air: Air in the atmosphere containing some water vapour (or moisture)

Dry air: Air that contains no water vapour.

- Water vapour in the air plays a major role in human comfort. Therefore it is an important consideration in air conditioning applications.
- Enthalpy of dry air: $h_{dry\ air} = c_p T$, where c_p is approximately 1.005 kJ/kg.K
- In air - conditioning processes we are concerned with the changes of enthalpy Δh

$$\Delta h_{dry\ air} = 1.005 \Delta T \text{ KJ/kg}$$

- Water vapour in air behaves as if it existed alone and obeys the ideal gas relation $Pv = RT$. Then the atmospheric air can be treated as an ideal gas-mixture:

Thus for atmospheric air

$P = P_a + P_v$, where P_a partial pressure of dry air and P_v partial pressure of water vapor(vapor pressure)

- We treat both the dry air and the water vapour as an ideal gas.
- We are most interested in temperatures from -10°C to 50°C.
- For atmospheric air, lower temperature at a fixed pressure and you hit the vapour saturation line → dew

fog

rain

Snow

- You get the same effect by lowering the pressure.
- The saturation temperature (dew point) goes down:
- $T_{dp} = P_{sat@p_v}$

Dry air	
$T, ^\circ\text{C}$	$c_p, \text{kJ/kg}\cdot^\circ\text{C}$
-10	1.0038
0	1.0041
10	1.0045
20	1.0049
30	1.0054
40	1.0059
50	1.0065

ENTHALPY APPROXIMATION (-10°C to 50°C)

Dry Air: $dh = c_p dT$

- $h_{dry,air} = C_p(T)$
- $h_{dry,air} = C_p \Delta T = 1.005 \frac{kJ}{kg \cdot ^\circ C} (T), T \text{ in } ^\circ C$
- $\Delta h_{dry air} = 1.005 \Delta T \text{ KJ/kg}$

Water vapour:

$h = h(T)$ since water vapour is an ideal gas

- $h_v(T, low P) \approx h_g(T)$
- $h_g(T) \approx 2500.9 + 1.82 T \text{ (kJ/Kg)}, T \text{ in } ^\circ C$

T, °C	Water vapor		
	$h_g, \text{kJ/kg}$		Difference, kJ/kg
	Table A-4	Eq. 15-4	
-10	2482.1	2482.7	-0.6
0	2500.9	2500.9	0.0
10	2519.2	2519.1	0.1
20	2537.4	2537.3	0.1
30	2555.6	2555.5	0.1
40	2573.5	2573.7	-0.2
50	2591.3	2591.9	-0.6

SPECIFIC HUMIDITY AND RELATIVE HUMIDITY

Absolute or specific humidity (humidity ratio)

- Water vapour mass to dry air mass or the mass of water vapour present in a unit mass of dry air.

$$\omega = \frac{m_v}{m_a}$$

- Using the Ideal gas equation ($PV = mRT$)

$$\omega = \frac{P_v V_v / R_v T_v}{P_a V_a / R_a T_a} = \frac{R_a}{R_v} \frac{P_v}{P_a} = 0.622 \frac{P_v}{P_a}, \quad \left(\frac{R_a}{R_v} = 0.287 / 0.4615 = 0.622 \right)$$

Using $P = P_a + P_v$

$$\omega = 0.622 \frac{P_v}{P - P_v}$$

Saturated air: The air saturated with moisture.

We can keep adding water vapour until we hit the saturated air point

→ air will not take any more moisture

It is thus logical to bound the amount of humidity between dry and saturated air.

$$P_v / \text{sat air} = P_g = P_{\text{sat press}} \text{ at temperature } T$$

$$\omega / \text{sat air} = 0.622 \frac{P_g}{P - P_g}$$

RELATIVE HUMIDITY (ϕ)

- Relative humidity is the ratio of the amount of moisture the air holds (m_v) and the maximum amount of moisture the air can hold at the same temperature (m_g)

$$\phi = \frac{m_v}{m_g}$$

m_v - moisture in air , m_g - maximum moisture air can hold

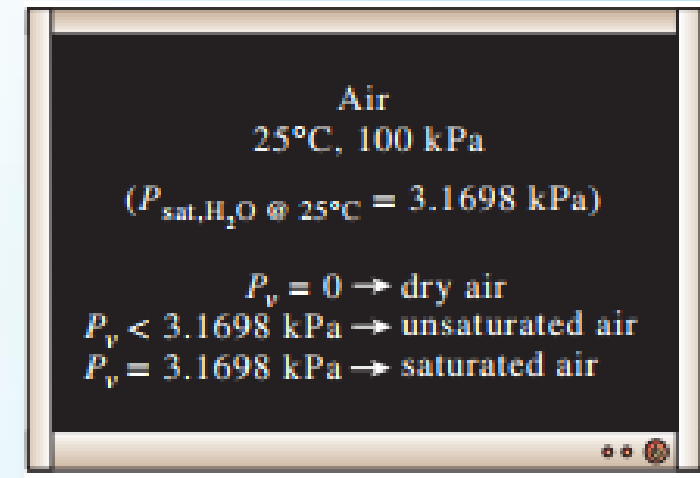
Using the ideal gas law

$$\phi = \frac{P_v V / R_v T}{P_g V / R_v T} = \frac{P_v}{P_g}, \quad P_g = P_{\text{sat}@T} \text{ from steam tables}$$

We can combine ϕ and ω :

$$\phi = \frac{\omega P}{(0.622 + \omega) P_g}; \quad \omega = \frac{0.622 \phi P_g}{P - \phi P_g}$$

What is the relative humidity of dry and saturated air?



ENTHALPY of Mixture or Atmospheric Air:

$$H = H_a + H_v = m_a h_a + m_v h_v$$

Dividing by m_a :

$$h = h_a + \frac{m_v}{m_a} h_v$$

$$h = h_a + \omega h_v$$

$$h_v \cong h_g$$

$$h = h_a + \omega h_g$$

Also note that the ordinary temperature of atmospheric air is frequently referred to as the **dry-bulb temperature** to differentiate it from other forms of temperatures that shall be discussed

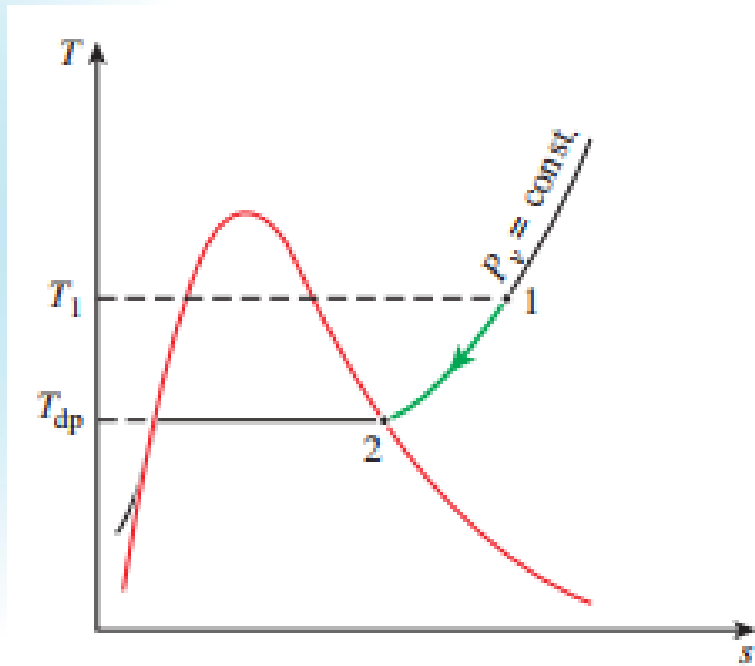
Example

1. A tank contains 15 kg of dry air and 0.17 kg of water vapor at 30°C and 100 kPa total pressure. Determine (a) the specific humidity, (b) the relative humidity, and (c) the volume of the tank.
2. A room contains air at 20°C and 98 kPa at a relative humidity of 85 percent. Determine (a) the partial pressure of dry air, (b) the specific humidity of the air, and (c) the enthalpy per unit mass of dry air.

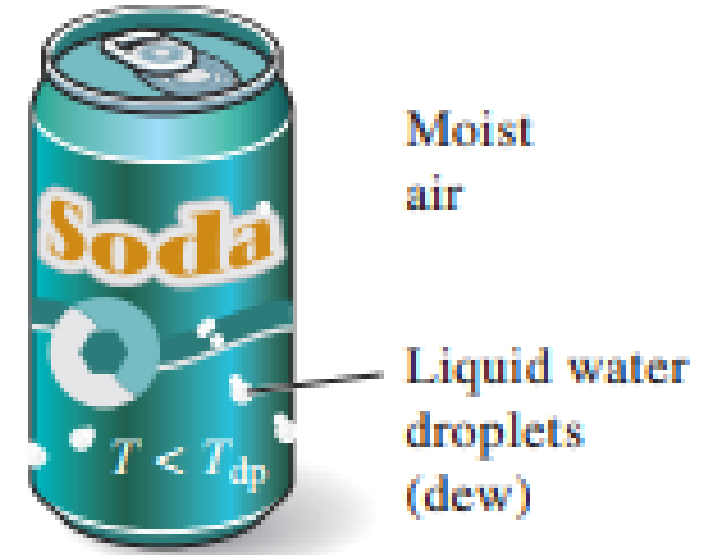
DEW-POINT TEMPERATURE

Summer mornings grass is wet but no rain? Moisture in air condenses on cool surface.

- During day water vaporizes
- At night, as temperature decreases, so does moisture capacity of air, that is, relative humidity goes up.
- Eventually, amount of moisture air can hold equals amount of moisture in air e.g. air becomes saturated with moisture.
- Any further decrease in temperature will lead to condensation ie. dew formation.
- **The dew-point temperature T_{dp}** is defined as the temperature below which the water vapor will condense out of the air.
- In other words, T_{dp} is the saturation temperature of water corresponding to the vapor pressure.
$$T_{dp} = T_{sat@pv}$$
- Further cooling, some vapour condenses out.
- Less vapor so vapor pressure decreases.



Constant-pressure cooling of moist air and dew –point temperature on the T-s diagram of water



When the temperature of a cold drink is below the dew-point temperature of the surrounding air, it “sweats”

Example 3

- A house contains air at 25°C with 65% relative. Will the any moisture condense if the window temperature is at 10°C ?

Example 4

- Atmospheric air at 30°C is passed through a cooler. Water begins to condense out of the air at 20°C . The $P_{\text{atm}} = 100 \text{ kpa}$.

Find ω , ϕ

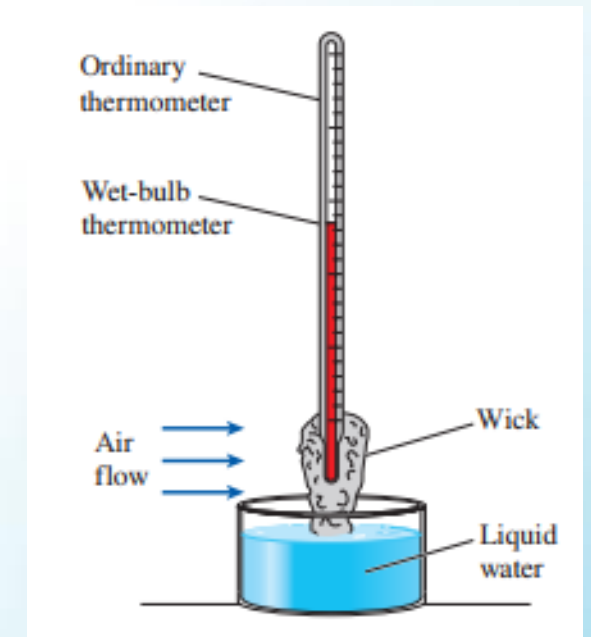
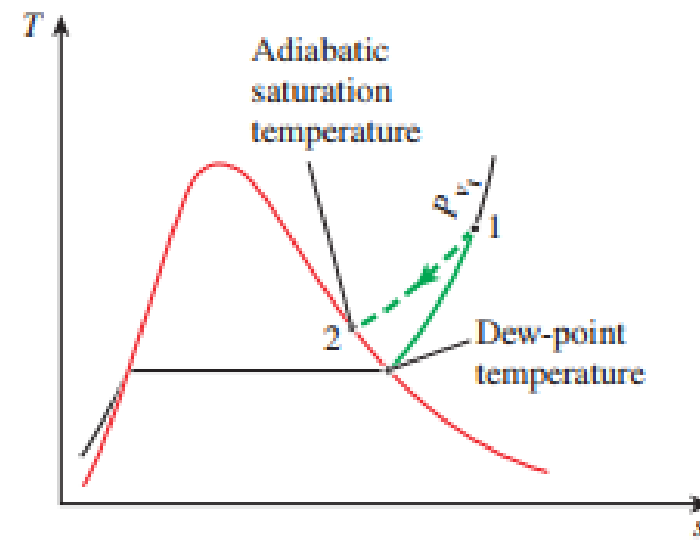
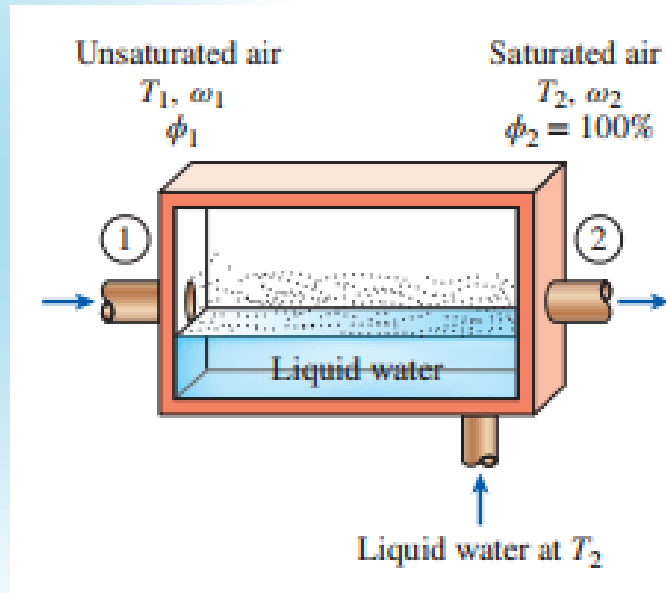
ADIABATIC SATURATION AND WET-BULB TEMPERATURE

Adiabatic Saturation Concept

- Definition: condition when air –water vapour mixture is brought to saturation by evaporating water into the air stream without heat transfer to surroundings.
- Process:
 - Unsaturated air passes over a water surface.
 - Evaporation occurs until equilibrium is reached.
- Key Assumptions: Adiabatic process (no external heat exchange).

Adiabatic Saturation Temperature

- The final temperature reached is called **the adiabatic saturation temperature**.
- It depends on:
 - Initial humidity of air
 - Pressure
 - Water temperature
- Used as a reference condition for humidity calculations.



The adiabatic saturation process and its representation on a T - s diagram of water

A simple arrangement to measure the wet-bulb temperature.

- The system consists of a long insulated channel that contains a pool of water.
- A steady stream of unsaturated air that has a specific humidity of ω_1 (unknown) and a temperature of T_1 is passed through this channel.
- As the air flows over the water, some water evaporates and mixes with the airstream.
- The moisture content of air increases during this process, and its temperature decreases, since part of the latent heat of vaporization of the water that evaporates comes from the air.
- If the channel is long enough, the airstream exits as saturated air ($\phi = 100$ percent) at temperature T_2 , which is called the **adiabatic saturation temperature**.
- If makeup water is supplied to the channel at the rate of evaporation at temperature T_2 , the adiabatic saturation process described above can be analyzed as a steady-flow process.
- The process involves no heat or work interactions, and the kinetic and potential energy changes can be neglected.
- Then the conservation of mass and conservation of energy relations for this two-inlet, one-exit steady-flow system reduce to the following:

Mass Balance:

$$\dot{m}_{a1} = \dot{m}_{a1} = \dot{m}_a \quad (\text{The mass flow rate of dry air remain constant})$$

$$\dot{m}_{w1} + \dot{m}_f = \dot{m}_{w2} \quad (\text{The mass flow rate of vapor in the air increases by an amount equal to the rate of evaporation } \dot{m}_f)$$

$$\text{Or } \dot{m}_a \omega_1 + \dot{m}_f = \dot{m}_a \omega_2$$

$$\text{Thus, } \dot{m}_f = \dot{m}_a (\omega_2 - \omega_1)$$

Energy Balance:

$$\dot{E}_{in} = \dot{E}_{out} \quad (\text{Since } \dot{Q} = 0 \text{ and } \dot{W} = 0)$$

$$\dot{m}_a h_1 + \dot{m}_f h_{f2} = \dot{m}_a h_2$$

Or

$$\dot{m}_a h_1 + \dot{m}_a (\omega_2 - \omega_1) h_{f2} = \dot{m}_a h_2$$

Dividing by $\dot{m}_a$ gives

$$h_1 + (\omega_2 - \omega_1) h_{f2} = h_2$$

Or

$$(c_p T_1 + \omega_1 h_{g1}) + (\omega_2 - \omega_1) h_{f2} = (c_p T_2 + \omega_2 h_{g2})$$

which yields

$$\omega_1 = \frac{c_p(T_2 - T_1) + \omega_2 h_{fg2}}{h_{g1} - h_{f2}}$$

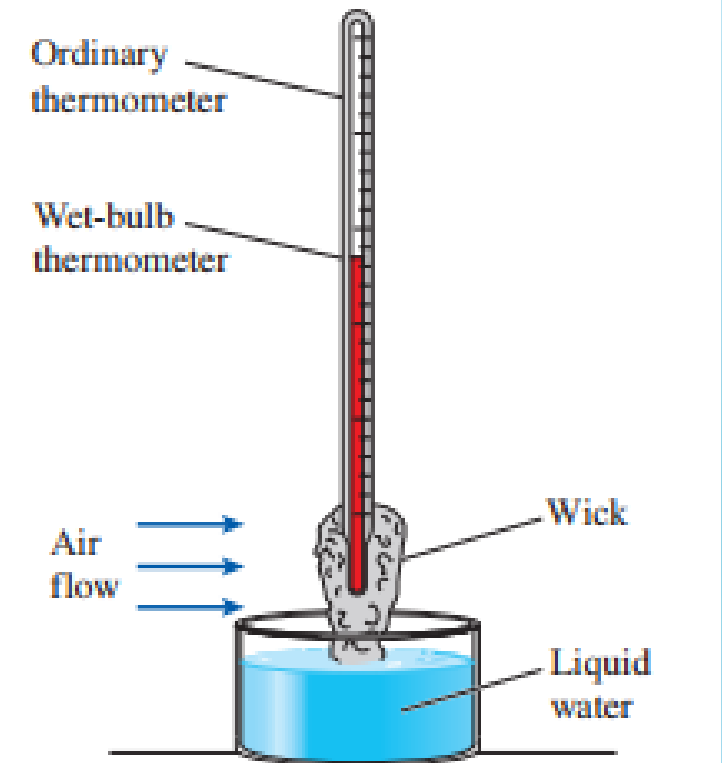
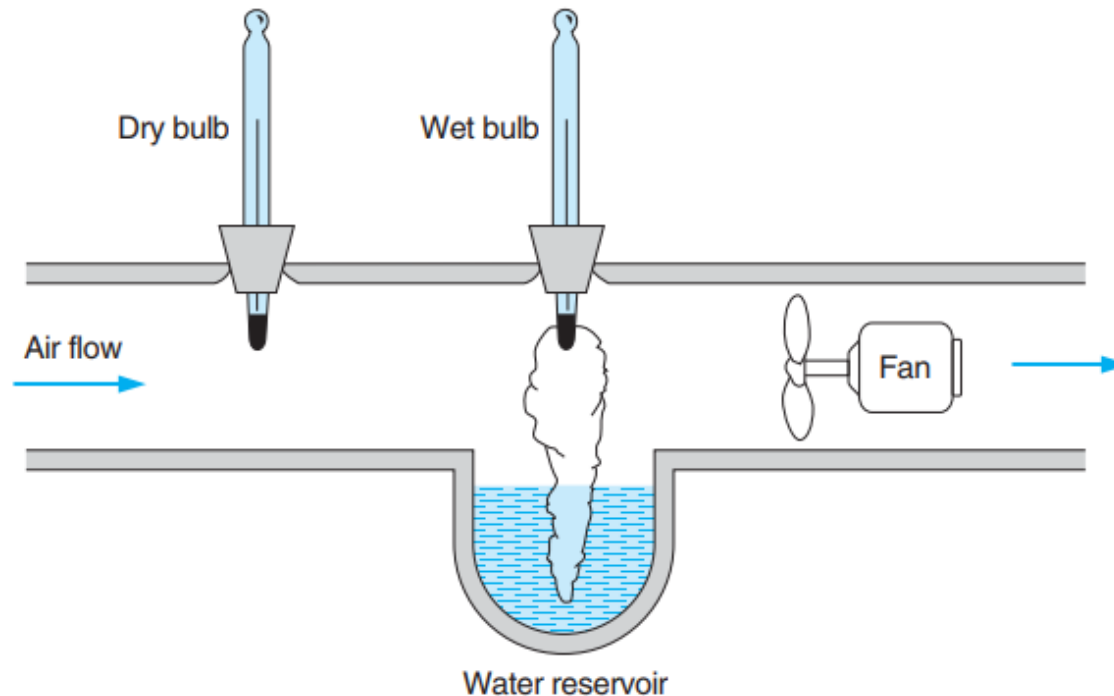
From previous equation,

$$\omega_2 = \frac{0.622 P_{g2}}{P_2 - P_{g2}}$$

Example 5:

The pressure of the mixture entering and leaving the adiabatic saturator is 0.1 MPa, the entering temperature is 30°C, and the temperature leaving is 20°C, which is the adiabatic saturation temperature. Calculate the humidity ratio and relative humidity of the air–water vapor mixture entering.

- The adiabatic saturation process discussed above provides a means of determining the absolute or relative humidity of air, but it requires a long channel or a spray mechanism to achieve saturation conditions at the exit.
- A more practical approach is to use a thermometer whose bulb is covered with a cotton wick saturated with water and to blow air over the wick, as shown in Fig. below.
- The temperature measured in this manner is called the **wet-bulb temperature** T_{wb} , and it is commonly used in air-conditioning applications.



$$\omega_1 = \frac{C_p(T_2 - T_1) + \omega_2 h_{fg2}}{h_{g1} - h_{f2}} \quad ; \quad \omega_2 = \frac{0.622 P_{g2}}{P - P_{g2}}$$

$$\phi_1 = \frac{\omega_1 P}{(0.622 + \omega_1) P_{g1}}$$

Example 6

The dry- and the wet-bulb temperatures of atmospheric air at 1 atm (101.325 kPa) pressure are measured with a sling psychrometer and determined to be 25°C and 15°C, respectively. Determine (a) the specific humidity, (b) the relative humidity, and (c) the enthalpy of the air.

The Psychrometric Chart

- Graphical tool used for determining air properties.
- The state of atmospheric air at a given pressure is defined by **two independent** properties.
- Air-conditioning analysis involves many repetitive calculations.
- Psychrometric charts simplify and visualize these calculations.

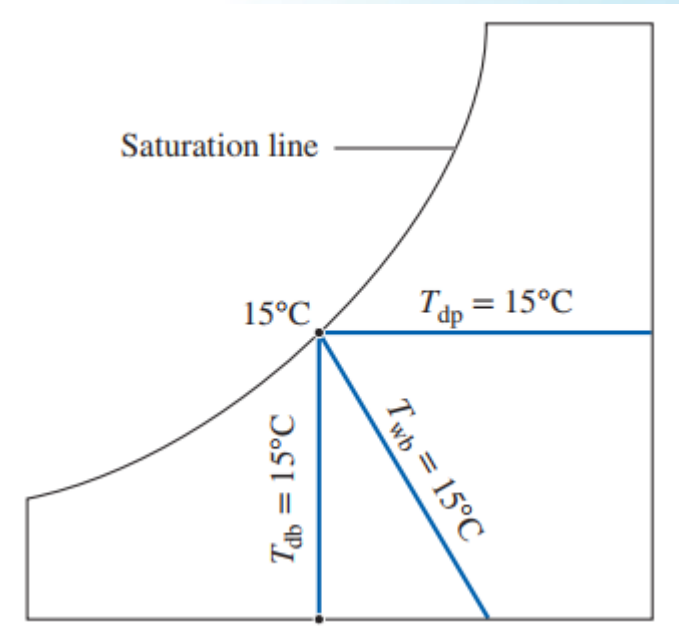
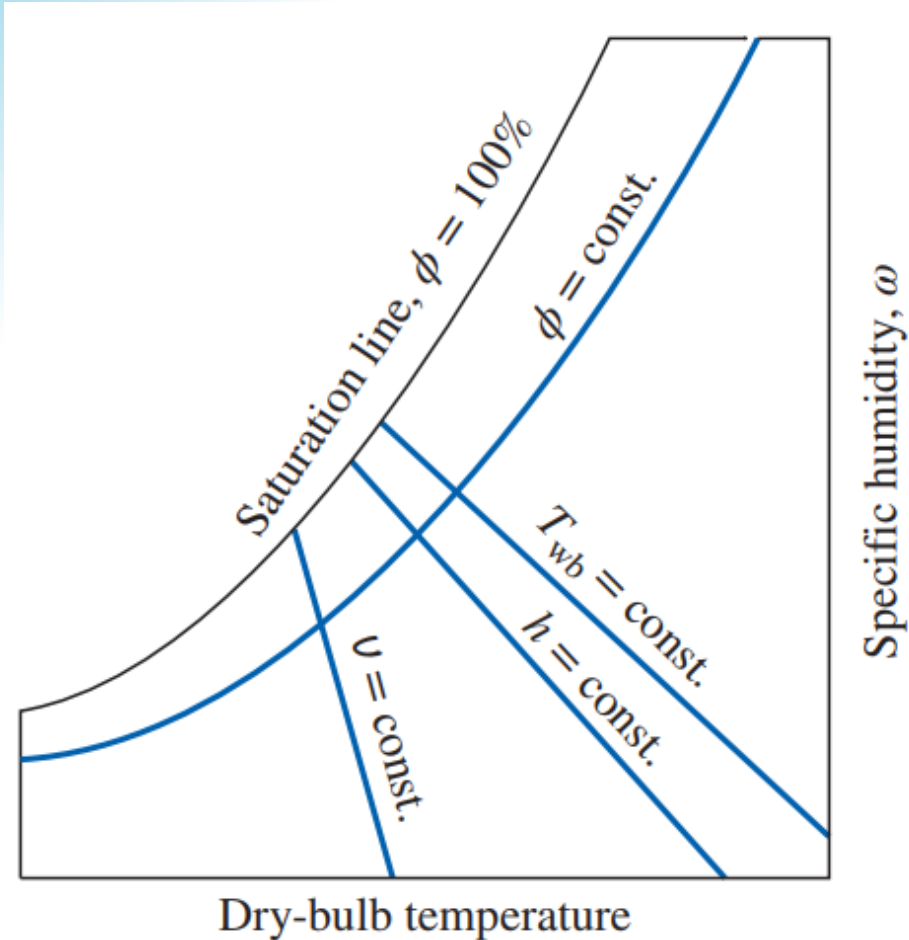
The basic features (lines and scales) of a psychrometric chart are:

- **Dry-Bulb (DB) Temperature:** Represented by vertical lines along the bottom (horizontal) axis, these lines indicate the air temperature as measured by a standard thermometer and increase from left to right.
- **Humidity Ratio (Moisture Content):** Indicated by horizontal lines, the humidity ratio measures the mass of water vapor per unit mass of dry air and is scaled on the vertical axis on the right side of the chart.

- **Relative Humidity (RH):** These are curved lines that sweep from the lower left to the upper right. The leftmost curve is the saturation curve, representing 100% relative humidity, where the air is fully saturated and condensation begins (dew point).
- **Wet-Bulb (WB) Temperature:** Shown as diagonal lines sloping downwards from left to right, the wet-bulb temperature is the temperature the air would reach if cooled to saturation by evaporation. These lines are almost parallel to the constant enthalpy lines.
- **Dew-Point (DP) Temperature:** The dew point temperature is read by following a horizontal line from a specific point on the chart to the saturation curve (100% RH line).
- **Enthalpy:** This represents the total heat energy (sensible and latent) of the air mixture. The scale is typically located outside the main body of the chart, with constant enthalpy lines running parallel to the wet-bulb temperature lines.
- **Specific Volume:** These lines indicate the volume occupied by a unit weight of dry air. They are steep, almost vertical diagonal lines.

Visualization of Processes

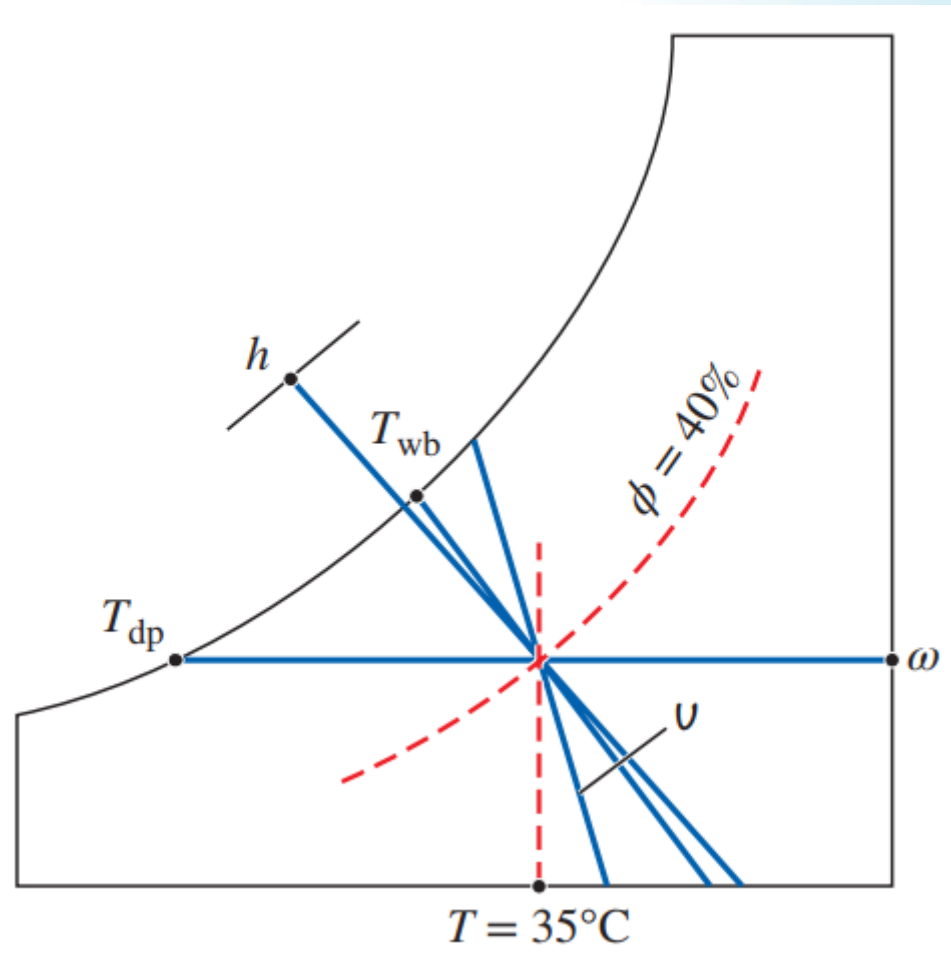
- Heating or cooling without moisture change appears as a horizontal line.
- Deviation from a horizontal line indicates humidification or dehumidification.



For saturated air, the dry-bulb, wetbulb, and dew-point temperatures are identical.

Example 7

Consider a room that contains air at 1 atm, 35°C , and 40 percent relative humidity. Using the psychrometric chart, determine (a) the specific humidity, (b) the enthalpy, (c) the wetbulb temperature, (d) the dew-point temperature, and (e) the specific volume of the air.



Solution

The relative humidity of air in a room is given. The specific humidity, enthalpy, wet-bulb temperature, dew-point temperature, and specific volume of the air are to be determined using the psychrometric chart.

Analysis At a given total pressure, the state of atmospheric air is completely specified by two independent properties such as the dry-bulb temperature and the relative humidity. Other properties are determined by directly reading their values at the specified state.

(a) The specific humidity is determined by drawing a horizontal line from the specified state to the right until it intersects with the v axis, as shown in the figure above.

$\omega = 0.0142 \text{ kg H}_2\text{O/kg dry air}$

(b) The enthalpy of air per unit mass of dry air is determined by drawing a line parallel to the $h = \text{constant}$ lines from the specific state until it intersects the enthalpy scale, giving

$h = 71.5 \text{ kJ/kg dry air}$

(c) The wet-bulb temperature is determined by drawing a line parallel to the $T_{wb} = \text{constant}$ lines from the specified state until it intersects the saturation line, giving

$T_{wb} = 24^\circ\text{C}$

(d) The dew-point temperature is determined by drawing a horizontal line from the specified state to the left until it intersects the saturation line, giving

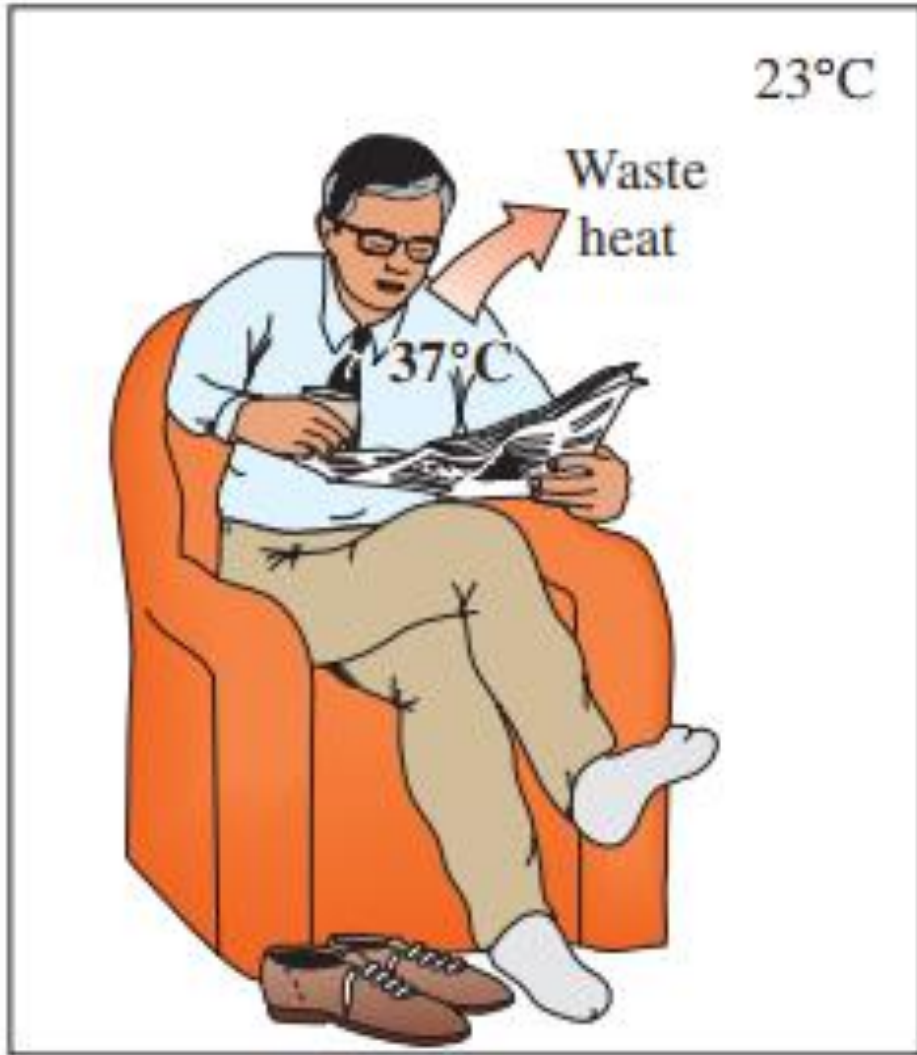
$T_{dp} = 19.4^\circ\text{C}$

(e) The specific volume per unit mass of dry air is determined by noting the distances between the specified state and the $v = \text{constant}$ lines on both sides of the point. The specific volume is determined by visual interpolation to be

$v = 0.893 \text{ m}^3/\text{kg dry air}$

Human Comfort and Air-Conditioning

- Human beings have an inherent weakness—they want to feel comfortable.
- Human beings want to feel comfortable i.e. Not too hot or too cold; not too sticky (humid) and not too dry.
- As engineers it is our duty to help people feel comfortable – besides it keeps us employed.
- We cannot change the weather but we can change the climate in a confined space by conditioning the air i.e. air conditioning.
- Human body can be viewed as a heat engine whose energy input is food.
- As with any other heat engine, the human body generates waste heat that must be rejected to the environment if the body is to continue operating.
- The rate of heat generation depends on the level of the activity.
- For an average adult male, it is about 87 W when sleeping, 115 W when resting or doing office work, 230 W when bowling, and 440 W when doing heavy physical work. The corresponding numbers for an adult female are about 15 percent less. This difference is due to the body size, not the body temperature



The deep-body temperature of a healthy person is maintained constant at about 37°C . A body will feel comfortable in environments in which it can dissipate this waste heat comfortably

A body feels comfortable when it can freely dissipate its waste heat, and no more

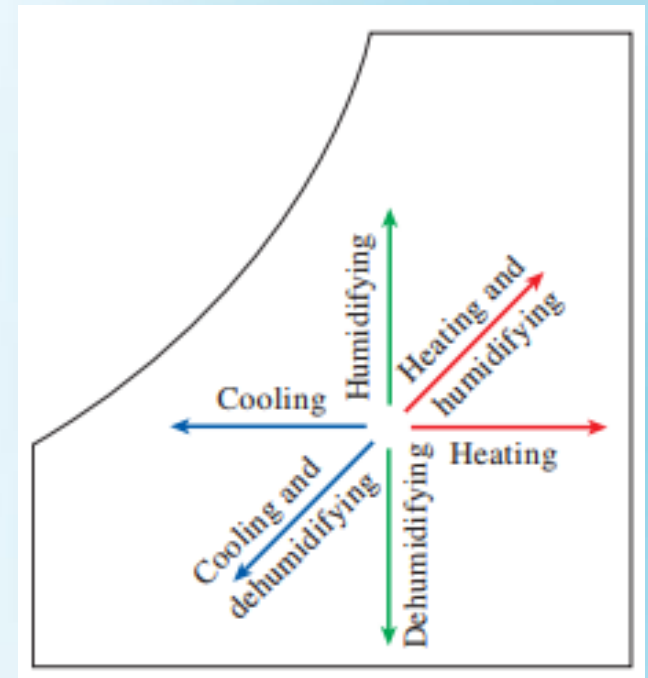
Factors Affecting Human Comfort

The comfort of the human body depends primarily on three factors: the (drybulb) temperature, relative humidity, and air motion.

- The temperature of the environment is the single most important index of comfort. Most people feel comfortable when the environment temperature is between 22°C and 27°C.
- The relative humidity also has a considerable effect on comfort since it affects the amount of heat a body can dissipate through evaporation. Relative humidity is a measure of air's ability to absorb more moisture. High relative humidity slows down heat rejection by evaporation, and low relative humidity speeds it up. Most people prefer a relative humidity of 40 to 60 percent.
- Air motion also plays an important role in human comfort. It removes the warm, moist air that builds up around the body and replaces it with fresh air. Therefore, air motion improves heat rejection by both convection and evaporation. Air motion should be strong enough to remove heat and moisture from the vicinity of the body, but gentle enough to be unnoticed. Most people feel comfortable at an airspeed of about 15 m/min

Air – Conditioning Processes

- Maintaining a living space or an industrial facility at the desired temperature and humidity requires some processes called air-conditioning processes.
- These processes include *simple heating* (raising the temperature), *simple cooling* (lowering the temperature), *humidifying* (adding moisture), and *dehumidifying* (removing moisture).
- Sometimes two or more of these processes are needed to bring the air to a desired temperature and humidity level.
- Air is commonly heated and humidified in winter and cooled and dehumidified in summer. Notice how these processes appear on the psychrometric chart.



Most air-conditioning processes can be modeled as steady-flow processes, and thus the *mass balance* relation $\dot{m}_{in} = \dot{m}_{out}$ can be expressed for *dry air* and *water* as

Mass balances for dry air: $\sum_{in} \dot{m}_a = \sum_{out} \dot{m}_a$

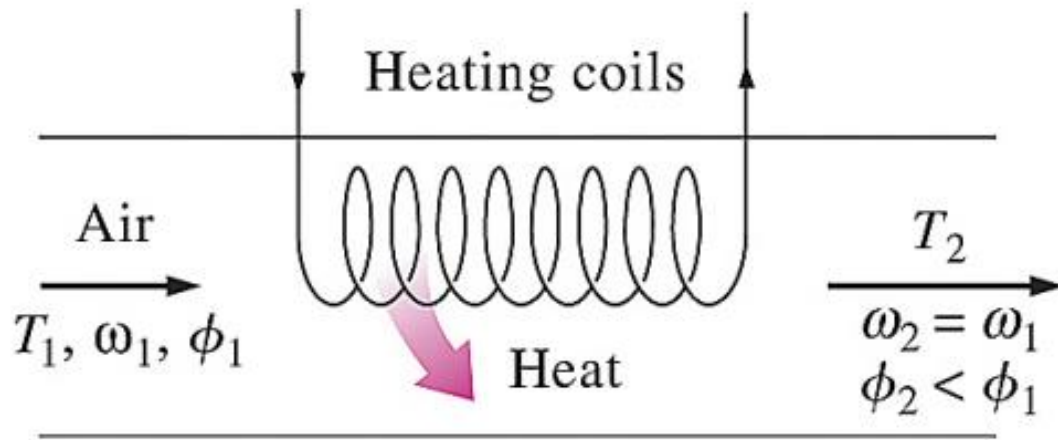
Mass balance for water: $\sum_{in} \dot{m}_w = \sum_{out} \dot{m}_w$

or

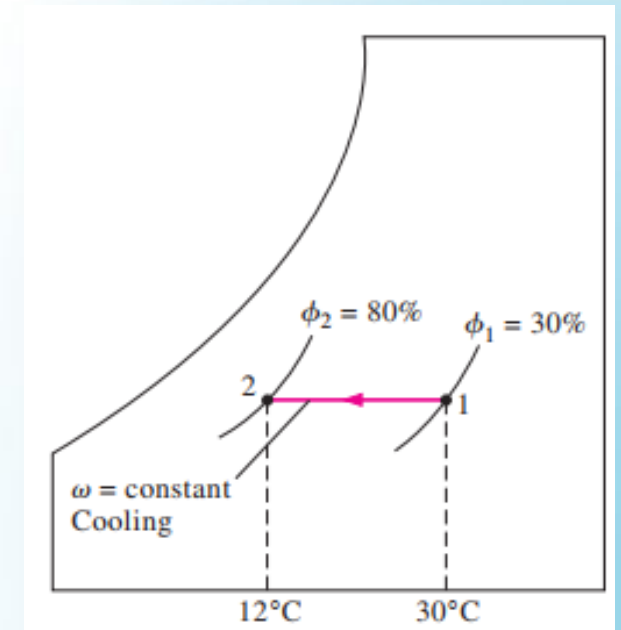
$$\sum_{in} \dot{m}_a \omega = \sum_{out} \dot{m}_a \omega$$

Simple Heating and Cooling ($\omega = \text{constant}$)

Many residential heating systems consist of a stove, a heat pump, or an electric resistance heater. The air in these systems is heated by circulating it through a duct that contains the tubing for the hot gases or the electric resistance wires, as shown in below.



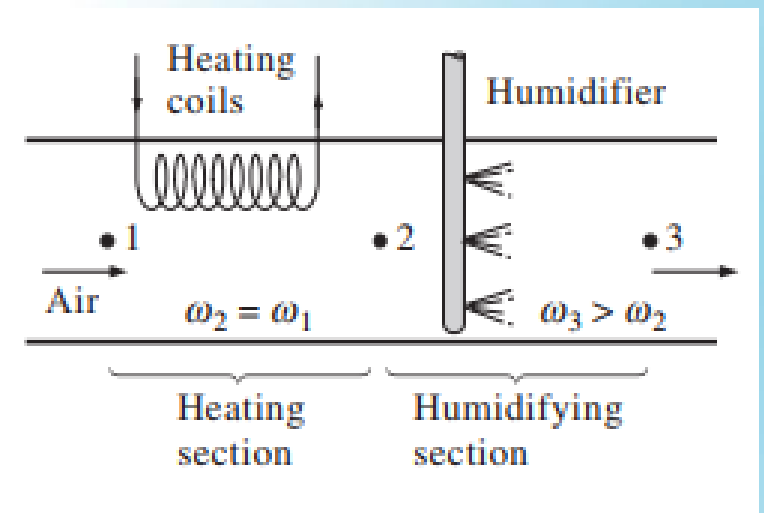
During simple heating, specific humidity remains constant, but relative humidity decreases.



During simple cooling, specific humidity remains constant, but relative humidity increases.

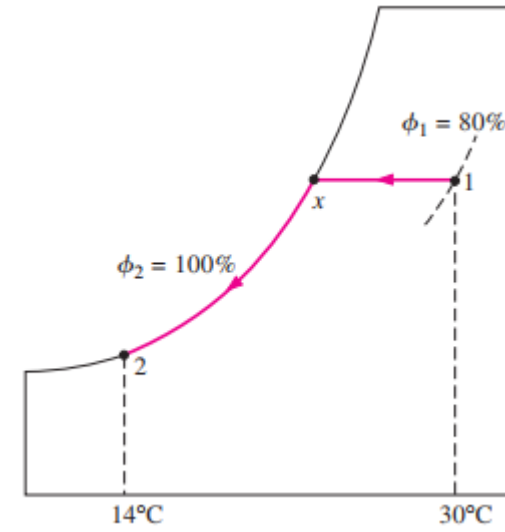
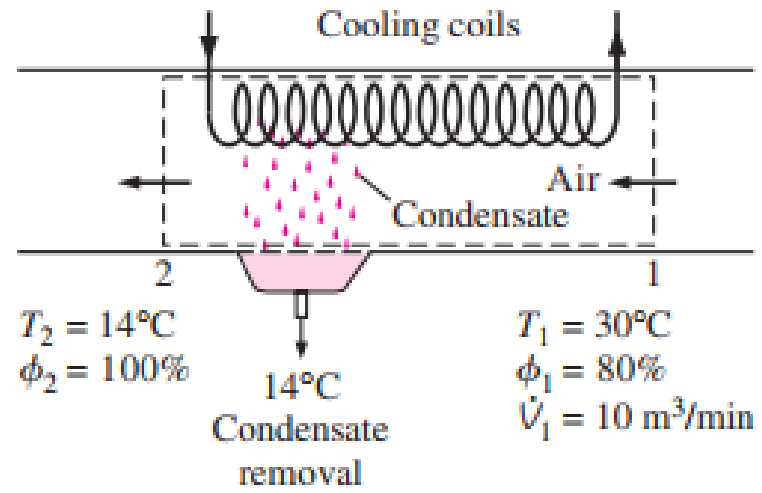
Heating with humidification

- Problems associated with the low relative humidity resulting from simple heating can be eliminated by humidifying the heated air.
- To maintain comfortable relative humidity, simple heating is typically accompanied with humidification, Fig. below. That is accomplished by passing the air through a humidifying section.
- If steam is used in the humidifier, we will have additional heating; thus $T_3 > T_2$. If water is sprayed in the humidifier section, part of the latent heat of vaporization comes from the air which results in the cooling of the air; thus $T_3 < T_2$.



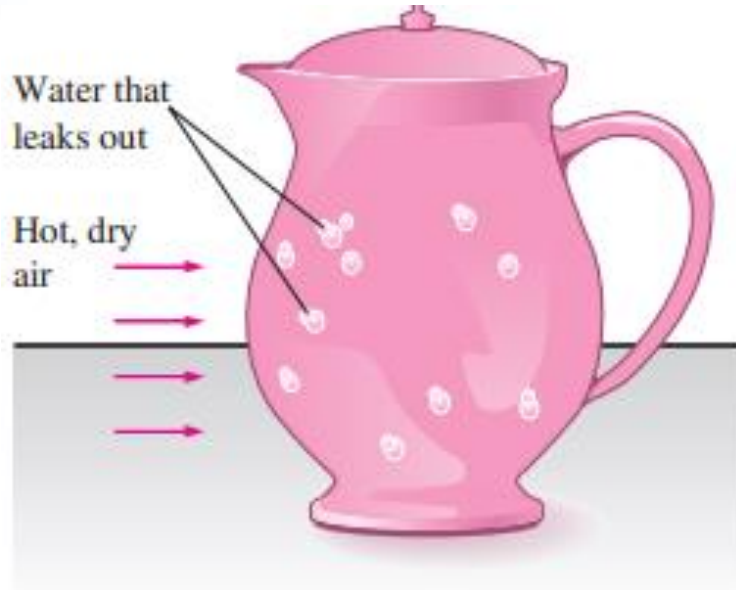
Cooling with Dehumidification

- The specific humidity of air remains constant during a simple cooling process, but its relative humidity increases.
- If the relative humidity reaches undesirably high levels, it may be necessary to remove some moisture from the air, that is, to dehumidify it.
- This requires cooling the air below its dewpoint temperature.
- To remove some moisture from the air, it should be cooled below its dew-point temperature.
- Passing through cooling coil, air temperature decreases and its relative humidity increases at constant specific humidity until air temperature reaches its dew point temperature.
- Any further cooling results in condensation of part of the moisture in the air.
- Note that air remains saturated during the entire condensation process.



Evaporative cooling

- Evaporative cooling is based on a simple principle: as water evaporates, the latent heat of vaporization is absorbed from the water body and the surrounding air.
- As a result, both water and air are cooled.
- The evaporative cooling is essentially identical to the adiabatic saturation process.
- Thus the evaporative cooling process follows a line of constant wetbulb temperature on the psychrometric chart.



Water in a porous jug left in an open, breezy area cools as a result of evaporative cooling.

Adiabatic Mixing of Airstreams

- Mixing processes normally involve no work interactions.
- Then the mass and energy balances for the adiabatic mixing of two airstreams reduce to:

Mass of dry air:

$$\dot{m}_{a_1} + \dot{m}_{a_2} = \dot{m}_{a_3}$$

Mass of water vapor:

$$\omega_1 \dot{m}_{a_1} + \omega_2 \dot{m}_{a_2} = \omega_3 \dot{m}_{a_3}$$

Energy:

$$\dot{m}_{a_1} h_1 + \dot{m}_{a_2} h_2 = \dot{m}_{a_3} h_3$$

Eliminating $\dot{m}_{a_3}$ the relations above, we obtain

$$\frac{\dot{m}_{a_1}}{\dot{m}_{a_2}} = \frac{\omega_2 - \omega_3}{\omega_3 - \omega_1} = \frac{h_2 - h_3}{h_3 - h_1}$$

